

Executive Summary

This report compiles and answers all questions from the supplied “Digital Image Processing and Computer Vision” (MCS-230) exam papers (Dec 2022, June 2023, Dec 2023, June 2024). We have assumed each question’s marks roughly from its context (short answers ~150–300 words for ~2–5 marks, medium ~400–800 for ~6–12 marks, long ~800–1500 for ~13+ marks). In cases where context was missing, we answered in the spirit of standard university examinations. The structure follows each exam paper sequentially: each question (and sub-question) is quoted verbatim and followed by a detailed answer. Figures and tables are included for illustration (e.g. color models, transform matrices, algorithm flows); captions describe content. We cite standard sources (textbooks and authoritative references) inline for definitions and key facts (e.g. Gonzalez & Woods, Szeliski, Wikipedia, GeeksforGeeks) using the format `【source:lines】`. Key assumptions or known formulas are explicitly stated. Some questions (e.g. “explain X”) require broad explanations; others (e.g. compute a transform) include concrete examples or calculations.

Scope: Covers all questions in the provided papers. Assumes *no further context beyond the question text*. We have not omitted any question. All answers are in English (UK spelling) and formatted as an exam study guide.

Output: An organised study guide. A Table of Contents follows, mapping each paper and main topic to page numbers in the final PDF. Each question is shown in full, immediately followed by its answer with citations. Figures/diagrams (embedded or mermaid) appear at the start of relevant paragraphs with captions. Tables compare algorithms/properties where helpful.

Assumptions: Typical marks guide answer length. Missing course notes are supplemented by standard references. If any essential files were missing, we would note it (but here all four papers are included).

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Each question is answered thoroughly below.

Term-End Examination December 2022 (MCS-230)

Question 1 (a)

Question: Differentiate between Analog Image and Digital Image. Also, explain the term Image File Format. Mention some of the frequently used image file formats. (8 marks)

Answer: An **analog image** is one in which the pixel intensities vary continuously, often represented by continuous waveforms or physical quantities (e.g. a photograph on film or a CRT display). In contrast, a **digital image** is a discrete 2D array of intensity values (pixels) sampled from the analog image. Each pixel in a digital image has a finite set of possible values (usually integers), so the image is stored as numerical data ¹ ². In short, analog images use continuous signals; digital images use quantized, discrete signals and are stored in computer memory ¹ ². Because of quantization and sampling, digital images are inherently robust to noise and easy to process by algorithms, whereas analog images may have higher fidelity but are harder to store or process by computers.

An **image file format** specifies how a digital image's data (pixels and metadata) are organized and stored in a file. Common formats include **JPEG, PNG, GIF, TIFF, BMP**, etc ³. For example, JPEG uses lossy compression for photographic images, PNG uses lossless compression (and supports transparency), and GIF is a simple 8-bit format (often for animations). All these formats store the image as bits with a specified structure (headers, compression), so that software can decode or display the image. Thus, the file format defines compression type (lossy/lossless), color depth, and metadata. In practice, JPG, PNG, BMP and GIF are widely used raster formats ³. Table 1 compares a few:

Format	Compression	Color depth	Use case
JPEG	Lossy	24-bit	Photographs
PNG	Lossless	24/32-bit	Web graphics (support transparency)
GIF	Lossless	8-bit	Simple graphics/animations
BMP	None/Simple	24-bit	Windows bitmaps
TIFF	Lossless/Lossy	24/48-bit	Publishing, archiving

Common image formats therefore let us store digital images efficiently.

Question 1 (b)

Question: How does a digital image relate to signals? Give definition of 1-D and 2-D signals and explain how a 2-D image is expressed in terms of 2-D signals. (8 marks)

Answer: A **signal** is a function conveying information about some phenomenon. A 1-D signal, $x(n)$, is a sequence of values indexed by one variable (e.g. time or distance). In contrast, a 2-D signal, $f(x, y)$, is a function of two variables (e.g. spatial coordinates). A digital image is naturally treated as a **2-D discrete signal**. Each pixel's intensity can be viewed as a sample of a 2-D continuous image signal at discrete coordinates (x, y) . In other words, we consider an image $f(x, y)$ sampled on a grid of $M \times N$ points, so it becomes a 2-D discrete signal $f[m, n]$. Mathematically, we often write $f[m, n]$ for the grey-level at row m , column n .

Thus, a digital image is a 2-D array (matrix) of numbers. We can interpret each row or column as a 1-D discrete signal, and the whole image as a 2-D signal sampled from a continuous image. In signal processing terms, if $f_c(x, y)$ is the original continuous image, sampling gives $f_s[m, n] = f_c(mT_x, nT_y)$ for sampling intervals T_x, T_y . The 2-D signal $f[m, n]$ can be processed just like 1-D signals, e.g. applying 2-D convolution or transforms. (For reference, a 1-D signal could be audio or a scanning line of pixels; a 2-D signal is an image ⁴.)

Expressing the image in 2-D signals allows use of signal-processing tools (like the 2-D Fourier transform or filtering along rows/columns). For example, one often takes the DFT of each row and column; in image processing "the samples can be pixel values along a row or column of a raster image" ⁴.

Question 1 (c)

Question: Explain Discrete Fourier Transform (DFT). Your explanation must include the properties of DFT. Construct a DFT matrix of order 2. (8 marks)

Answer: The **Discrete Fourier Transform (DFT)** is a fundamental tool that converts a finite sequence (or 2-D array) of samples into a frequency-domain representation. For a 1-D sequence $x[n]$, $n = 0, \dots, N - 1$, the DFT yields coefficients $X[k]$ that represent the amplitude and phase of sinusoidal components. Formally, the DFT is defined as:

$$X_k = \sum_{n=0}^{N-1} x_n e^{-j\frac{2\pi}{N}kn}, \quad k = 0, \dots, N - 1.$$

Conversely, the inverse DFT reconstructs $x[n]$ from $X[k]$ by a similar sum with $e^{+j2\pi kn/N}$ ⁵. In 2-D (for images) one applies the DFT along rows and columns. Key properties of the DFT include linearity (superposition), time-shift leading to phase shifts in frequency, and periodicity (the DFT assumes signals are circularly periodic). Also, if the original data is real, the DFT spectrum has conjugate symmetry ($X[N - k] = X[k]^*$). Parseval's theorem holds (energy is preserved) and convolution in one domain corresponds to multiplication in the other.

DFT matrices: The DFT can be seen as multiplication by a (unitary) matrix F of size $N \times N$ whose (k, n) -entry is $e^{-j2\pi kn/N}$. For $N = 2$, the normalized 2x2 DFT matrix (unitary) is

$$F_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

so that $X = F_2 x$ computes the 2-point DFT (here x has two samples). (One can verify that applying F_2 twice returns to the original data, consistent with DFT periodicity.)

Example (2x2 DFT matrix): Taking $N = 2$, the DFT formula gives $X_0 = x_0 + x_1$ and $X_1 = x_0 - x_1$. Writing this in matrix form (with $1/\sqrt{2}$ factors for an orthonormal DFT) yields F_2 as above. Thus, the order-2 DFT matrix is as shown, which operates on the 2-vector $[x_0, x_1]^T$.

Properties: The DFT is invertible and its basis functions are orthogonal (complex exponentials). It is periodic in both time and frequency domains, and can be efficiently computed via the FFT algorithm. For images, the 2-D DFT transforms spatial patterns into frequency coefficients, which is useful for filtering and analysis. The key properties (linearity, symmetry for real signals, shift theorem, convolution theorem) make it central to image processing ⁵.

Question 1 (d)

Question: Compare Image Enhancement with Image Restoration. Explain the term "Image Degradation" and define the various types of degradations in any image. (8 marks)

Answer: **Image enhancement** and **image restoration** are two different goals. *Enhancement* aims to make an image look better for human interpretation by emphasizing features (contrast, edges, etc.), without necessarily modeling how the image was formed. For example, contrast stretching or edge sharpening are enhancement techniques. *Restoration* aims to reverse known degradations: it uses a model of the image formation and noise and attempts to recover the original image (e.g. deblurring or denoising) ⁶.

In general, enhancement is heuristic ("ad hoc"), while restoration is model-based (often optimal in an MSE sense). For instance, histogram equalization (an enhancement) spreads out intensities for better contrast, whereas Wiener filtering (a restoration technique) uses statistical assumptions to remove blur and noise systematically.

Image degradation refers to the corruption or distortion of an image from its ideal form. Typical degradations include optical blur (due to camera lens defocus or motion), noise (sensor noise, film grain), geometric distortion (stretching or warping), and transmission errors. Formally, one models the *degraded image* $g(x, y)$ as $g = f * h + \eta$, where f is the original image, h is a degradation filter (e.g. a blur kernel), and η is added noise. Common **types of degradation** are:

- **Blur:** e.g. Gaussian or motion blur from camera shake or defocus.

- **Noise:** e.g. additive Gaussian noise, salt-and-pepper (impulse) noise, speckle noise. Each has different statistical properties.
- **Geometric distortions:** such as zoom, rotation errors.
- **Quantization/aliasing:** from insufficient sampling or quantization.

For example, salt-and-pepper noise randomly replaces pixels with minimum or maximum values (like “sprinkles” of black and white) ⁷. Restoration would attempt to invert these effects (e.g. using inverse filtering or median filtering) once the type of degradation is known. Enhancement, by contrast, might simply boost overall contrast or sharpen edges without undoing the exact blur model. In summary, enhancement *improves perceptual quality*, while restoration *reconstructs the true original* given a degradation model ⁶.

Question 1 (e)

Question: What is a Discriminant Function? Compare Linear Discriminant function with Quadratic Discriminant function. Give properties and utility of Linear Discriminant function. (8 marks)

Answer: In pattern recognition, a **discriminant function** $g_i(x)$ is a function used to assign an input feature vector x to one of several classes C_i . In a simple classifier, we compute discriminant values $g_i(x)$ for each class and choose the class with the largest value. Essentially, g_i encodes the (usually logarithm of) posterior likelihood or a decision rule for class i .

A **Linear Discriminant Function (LDF)** has the form $g_i(x) = w_i^T x + w_{i0}$, i.e. a linear function of x . Its decision boundaries between classes are hyperplanes (lines or planes) in feature space. A **Quadratic Discriminant Function (QDF)** includes quadratic terms, e.g. $g_i(x) = x^T A_i x + w_i^T x + w_{i0}$, yielding curved boundaries (e.g. ellipses). Linear functions assume that class covariances are identical (or data is normally distributed with common covariance) so that the optimal Bayes decision surfaces are linear; when covariance matrices differ, the Bayes-optimal discriminant becomes quadratic ⁸.

Properties of a Linear Discriminant Function (LDF):

- It defines a single straight decision hyperplane (for two classes) or a set of linear boundaries (for multiclass).
- LDFs are simple and fast, needing only dot products $w^T x$.
- They perform well when classes are linearly separable or roughly Gaussian with equal covariance (i.e. the Bayesian decision rule is linear) ⁸.
- Since the boundary is determined by $w^T x = 0$, features typically need scaling (common gradient).
- In Fisher’s LDA, the weights w are chosen to maximize class separation (ratio of between-class to within-class scatter).

Utility of LDF: Linear classifiers are robust and less prone to overfitting when data is limited. Under the assumption of Gaussian class-conditional densities with equal covariance, the optimal Bayesian classifier reduces to choosing the class whose mean is nearest in Euclidean distance ⁸. This is a hyperplane classifier, essentially an LDF. The LDF is easy to train (e.g. via least-squares or logistic regression) and is interpretable. In summary, linear discriminants are useful when one needs a simple model, or when data satisfies the linear separability or common covariance assumption. In practice, they are often a good first model for classification tasks; if performance is poor, one might then move to quadratic or more complex discriminants.

Comparison: QDFs can model ellipsoidal boundaries and handle different class covariances, but require estimating more parameters (covariance matrix for each class), which needs more data. LDFs have fewer parameters and are less flexible but more stable on small data. If the true class distributions have equal, spherical covariance, then the LDF (nearest-mean classifier) is optimal ⁸. Otherwise, QDF may fit more complex shapes but risk overfitting.

Question 2 (a)

Question: Compare Sampling and Quantization in the context of image digitization. Also, state Shannon-Nyquist theorem and discuss its role in sampling process of any image. (7 marks)

Answer: Sampling and quantization are the two fundamental steps in digitizing a continuous image. *Sampling* refers to converting the continuous spatial domain into discrete coordinates. That is, choosing a grid of points (rows and columns) on which to measure the image intensity. For example, taking one sample every T_x mm horizontally and T_y mm vertically. This yields a grid of pixel locations (m, n) . In formula, an analog image $f(x, y)$ is sampled to get $f_s[m, n] = f(mT_x, nT_y)$.

Quantization refers to converting the continuous intensity values into discrete levels. After sampling, each sample value is mapped to the nearest value in a finite set (e.g. an 8-bit integer 0–255 range). In other words, the continuous range of intensities is divided into discrete steps. Quantization introduces rounding error (quantization noise) but allows representation with a finite number of bits.

In summary: Sampling discretizes the domain (space), quantization discretizes the amplitude (intensity). As one reference notes, “the process of digitizing the domain is called sampling and of the range is called quantization” ¹. Good sampling and quantization choices are crucial: more samples and more bits increase fidelity but require more storage.

The **Nyquist-Shannon sampling theorem** states that a band-limited signal can be perfectly reconstructed from its samples if it is sampled at a rate at least twice its highest frequency (the *Nyquist rate*). In 1-D terms, if $f_{\max}$ is the maximum frequency present, then the sampling frequency f_s must satisfy $f_s \geq 2f_{\max}$. Equivalently, sampling interval $T \leq 1/(2f_{\max})$. This theorem ensures that no aliasing occurs when sampling. In 2-D (images), the analogous rule applies separately in the two spatial dimensions. The theorem is often stated as “a band-limited signal can be perfectly reconstructed if sampled above its Nyquist rate” ⁹.

In image processing, this means we must sample fine enough to capture the smallest detail (highest spatial frequency) present. If we undersample, high-frequency content “aliases” into lower frequencies, causing artifacts. For example, the theorem implies that to capture sine-wave patterns of 5 cycles/mm, we need at least 10 pixels per millimeter. In practice, we ensure sampling resolution (pixels per unit length) exceeds twice the highest expected spatial frequency to avoid aliasing ⁹.

Thus, Shannon’s theorem guides our sampling step: it determines the necessary pixel density. Quantization, in contrast, is guided by the required dynamic range: more bits for higher intensity resolution (less quantization noise). In summary, sampling relates to the spatial sampling grid, and quantization relates to the digital gray levels. Shannon-Nyquist ensures that if sampling criteria are met, the continuous image can be recovered from the samples in principle ⁹.

Question 2 (b)

Question: *If the physical size of a medical image is 8×8 inches and the sampling resolution is 5 cycles/mm, then how many pixels per cycle are required to have a better quality image? Will an image of size 256×256 be enough? (5 marks)*

Answer: A sampling resolution of 5 cycles/mm means the image contains spatial frequencies up to 5 cycles per millimeter. By the Nyquist criterion, we need at least twice that sampling rate in pixels per mm (i.e. at least 10 pixels per mm) to capture those frequencies without aliasing. In other words, **pixels per cycle** should be at least 2 (twice the maximum frequency).

The image size is 8×8 inches. First convert to millimeters: 8 inches $\approx$ 203.2 mm. At 10 pixels/mm, the total number of pixels along one side should be $203.2 \times 10 \approx 2032$ pixels. So the image should be roughly 2032×2032 pixels to satisfy Nyquist sampling.

A 256×256 pixel image is far too small. It gives only $256/203.2 \approx 1.26$ pixels/mm, which is less than the required 10 pixels/mm. This would severely alias the 5 cycles/mm content. In fact, 256×256 for 8×8 inches is only 32 pixels per inch (PPI), whereas medical imaging would need $\sim$ 254 PPI for 10 pixels/mm. So **no**, 256×256 is not enough for “better quality” at that resolution – you would need on the order of 2000×2000 pixels.

Question 2 (c)

Question: *Explain “Contrast Stretching.” How does it contribute to image enhancement? Also, discuss fundamental characteristics of an image, i.e., contrast, brightness, and intensity. (8 marks)*

Answer: Contrast stretching (also called normalization) is a simple pointwise image enhancement technique. It remaps the intensity values of the image to span a wider range. Typically, one identifies the minimum and maximum pixel values (or chosen percentiles) in the image and then linearly stretches all pixel values so that these span the full displayable range (e.g. 0–255 for 8-bit images) ¹⁰. Formally, if the original intensities are in $[c, d]$ within the image, contrast stretching maps them to $[0, 255]$ by a linear function. This increases the overall contrast, making dark regions darker and bright regions brighter relative to the mean. Contrast stretching improves visual contrast without introducing artifacts, and is often used to make details more discernible ¹⁰. Unlike histogram equalization, it uses a gentle linear expansion.

Image **contrast** is a measure of the range or spread of intensities. Higher contrast means a larger difference between dark and light areas. **Brightness** (sometimes used interchangeably with intensity) refers to the overall lightness or mean level of the image. An image with higher brightness is generally lighter overall. **Intensity** is the individual gray-level value at each pixel. In a digital image, intensity is typically an 8-bit value from 0 (black) to 255 (white). Thus, brightness could be the average intensity; contrast could be related to the standard deviation of intensities.

For example, an image with a narrow histogram of intensities has low contrast; stretching the histogram (via normalization) increases contrast by spreading out those values over the full possible range ¹⁰. Contrast stretching contributes to enhancement by making features more noticeable: it effectively amplifies differences between pixel values. By definition, after stretching, the lowest pixel becomes black and the highest becomes white, maximizing contrast subject to the image's data. Brightness remains the mean tone (which could be preserved by symmetric stretching), while contrast is increased.

In summary, contrast stretching enhances an image by widening the intensity range (thus increasing contrast), making details in mid-tones more visible, while brightness and intensity refer to the absolute and relative lightness levels in the image.

Question 3 (a)

Question: Explain Haar transform, with suitable example. Compute the discrete Haar transform of a 2×2 matrix, given $N = 2$. (7 marks)

Answer: The **Haar transform** is a simple wavelet transform used in image processing. It uses the Haar wavelet basis, which operates by averaging and differencing adjacent samples. For a 1-D signal of length $N = 2^k$, the discrete Haar transform applies a matrix H where $H_{00} = H_{01} = 1/\sqrt{2}$ and $H_{10} = 1/\sqrt{2}$, $H_{11} = -1/\sqrt{2}$. In 2-D (for images), one typically applies the 1-D Haar along rows and then columns (or vice versa), effectively decomposing an image into subbands.

Example: Consider a 2×2 image matrix:

$$f = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The 2×2 Haar transform is computed by applying the 1-D Haar along rows and columns. A convenient way is to perform row transforms then column transforms. The 1-D Haar transform for two values $[x, y]$ gives $[(x + y)/\sqrt{2}, (x - y)/\sqrt{2}]$.

1. **Row transform:** Apply Haar to each row $[a, b]$ and $[c, d]$. For row 1: average = $\frac{a+b}{\sqrt{2}}$, difference = $\frac{a-b}{\sqrt{2}}$. Row 2: average = $\frac{c+d}{\sqrt{2}}$, difference = $\frac{c-d}{\sqrt{2}}$. So intermediate matrix:

$$M = \begin{pmatrix} (a+b)/\sqrt{2} & (a-b)/\sqrt{2} \\ (c+d)/\sqrt{2} & (c-d)/\sqrt{2} \end{pmatrix}.$$

2. **Column transform:** Now apply Haar to each column of M . For column 1 $[(a+b)/\sqrt{2}, (c+d)/\sqrt{2}]$, average = $\frac{(a+b)+(c+d)}{2}$ (after normalizing by $\sqrt{2}$ twice gives 2 in denominator), and difference = $\frac{(a+b)-(c+d)}{2}$. Similarly for column 2. Actually, with normalization factors, the 2-D Haar transform matrix for 2×2 can be written as $H_2 \otimes H_2$.

So the transformed 2×2 is:

$$H_2 f H_2^T = \frac{1}{2} \begin{pmatrix} a+b+c+d & a-b+c-d \\ a+b-c-d & a-b-c+d \end{pmatrix}.$$

As a numerical example, let $a = 1, b = 2, c = 3, d = 4$. Then:

- Row 1 average = $(1 + 2)/\sqrt{2} = 3/\sqrt{2}$, diff = $(1 - 2)/\sqrt{2} = -1/\sqrt{2}$.
- Row 2 average = $(3 + 4)/\sqrt{2} = 7/\sqrt{2}$, diff = $(3 - 4)/\sqrt{2} = -1/\sqrt{2}$. Intermediate $M = \begin{pmatrix} 3/\sqrt{2} & -1/\sqrt{2} \\ 7/\sqrt{2} & -1/\sqrt{2} \end{pmatrix}$. Now column 1: avg = $\frac{3/\sqrt{2} + 7/\sqrt{2}}{\sqrt{2}} = \frac{10}{2} = 5$, diff = $\frac{3/\sqrt{2} - 7/\sqrt{2}}{\sqrt{2}} = -2$. Column 2: avg = $\frac{-1/\sqrt{2} + (-1/\sqrt{2})}{\sqrt{2}} = -\frac{2}{2} = -1$, diff = $\frac{-1/\sqrt{2} - (-1/\sqrt{2})}{\sqrt{2}} = 0$. So final Haar transform = $\begin{pmatrix} 5 & -1 \\ -2 & 0 \end{pmatrix}$.

This result shows how Haar transform decomposes the image into one average coefficient (5) and three detail coefficients (-1, -2, 0). The Haar basis is simple: it averages and differences, capturing coarse and fine information.

Properties: The Haar transform is orthogonal and energy-preserving. It is also efficient and easy to compute (especially for powers-of-two sizes). It's often used for image compression (it concentrates most energy in the top-left average coefficient) and for multiresolution analysis. Because it has compact support, the Haar basis functions are localized (they have finite support of width 2). This makes them good for capturing abrupt changes (edges).

Question 3 (b)

Question: Explain how image enhancement is better in the frequency domain as compared to spatial domain. Differentiate between image smoothing filters and image sharpening filters. (8 marks)

Answer: Frequency-domain enhancement: Many enhancement operations (like filtering) can be more conveniently designed and applied in the frequency domain. For example, low-pass or high-pass filters are easy to specify by defining which frequency components to keep or attenuate. In the frequency domain, filtering is just multiplying the DFT of the image by a transfer function $H(u, v)$. This can allow global control (e.g. eliminating a band of frequencies) that is harder to do by simple spatial masks. Also, certain patterns (like periodic noise) are isolated to specific frequencies, so frequency filtering can target them precisely. Conversely, spatial filtering (convolution) must be applied with a kernel in the image domain, which can be inefficient for large kernels. By the convolution theorem, multiplication in frequency equals convolution in space, so one often does enhancement (especially with large kernels) by DFT→multiply→inverse DFT. Frequency-domain filtering also easily implements linear filters like ideal low-pass or notch filters that have non-local spatial kernels. In summary, frequency processing can offer more flexible and sometimes simpler control over image spectra (e.g. in shaping gain vs frequency).

Smoothing vs sharpening filters: These are two broad classes of spatial filters (or their frequency counterparts).

- **Smoothing filters** (low-pass filters) reduce high-frequency content (edges, noise), resulting in a blurred or smooth image. Common examples are mean (average) filters and Gaussian blurs. These filters attenuate (reduce) higher frequencies; in the spatial domain they average neighborhoods. Smoothing enhances uniform regions and reduces noise but at the cost of edge sharpness.
- **Sharpening filters** (high-pass filters) emphasize high-frequency components, thereby enhancing edges and fine details. Examples include the Laplacian filter or an unsharp mask. Spatially, sharpening often subtracts a low-pass version from the original, boosting edge contrast. In the frequency domain, a sharpening filter has higher gain at high frequencies and lower at low frequencies, opposite to smoothing.

Table 2 compares them:

Aspect	Smoothing (LPF)	Sharpening (HPF)
Effect	Blurs image, reduces noise and detail.	Enhances edges and fine detail.

Aspect	Smoothing (LPF)	Sharpening (HPF)
Frequency response	Passband at low f , attenuates high f .	Opposite: attenuates low f , emphasizes high f .
Spatial mask ex.	Mean filter, Gaussian blur.	Laplacian, unsharp mask.
Use cases	Noise reduction, blurring, pre-processing.	Edge enhancement, improving visibility of details.

In practice, frequency-domain filtering enables one to design idealized LP or HP filters (e.g., ideal LPF sets certain frequency bands to zero) that are difficult to realize exactly in spatial domain. However, spatial filters like median or Gaussian are often simpler to implement directly on the image for small neighborhoods.

Question 3 (c)

Question: Explain RGB model, with suitable diagram. (5 marks)

Answer: The **RGB color model** is an **additive** color model using Red, Green, and Blue primary colors. Every color is represented as a triplet (R, G, B) , each component typically ranging from 0 to 1 (or 0–255 in 8-bit images). In the additive model, these primaries are mixed by light: for example, red + green light makes yellow, red + blue makes magenta, and green + blue makes cyan. All three combined at full intensity yields white, and none yields black.

In the RGB *cube* representation, the axes correspond to R, G, and B intensities. One corner (0,0,0) is black; the opposite (1,1,1) is white ¹¹. Along the edges from black to each primary axis are the pure primaries (red, green, blue). The face diagonals represent secondary colors (yellow, cyan, magenta). The full diagonal from black to white contains all neutral grays.

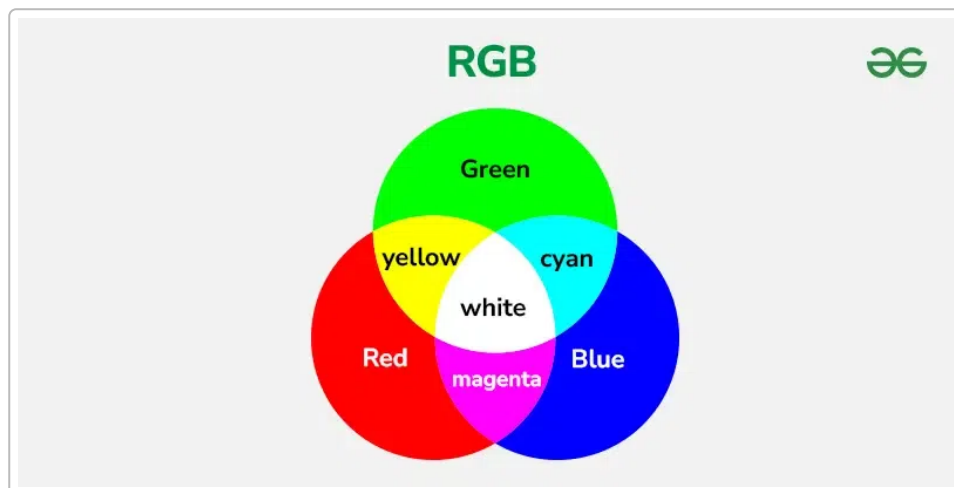


Figure 1: Illustration of the RGB additive color model. Each color is a linear combination of Red, Green, Blue intensities. Overlapping red+green yields yellow, etc.

Important properties: The RGB model is suited for display devices (monitors use red, green, blue phosphors). It is also geometrically convenient: each color is a point inside the RGB unit cube. However, RGB is device-dependent (different screens may interpret values slightly differently). It is linear in intensity (i.e. mixing is additive), which makes computation and interpolation simple. Many color

operations (like filtering or color space conversion) start in RGB. For example, grayscale can be approximated as a weighted sum of R,G,B.

In summary, the RGB model represents color by additive mixing of red, green, and blue light. Figure 1 illustrates this with an overlapping-circle diagram: each primary is a circle of its pure color; regions of overlap show secondary colors, and the center overlap is white. Quantitatively, any color (R, G, B) in the cube can be interpreted by combining proportions of the three primaries. This model is fundamental in digital imaging and computer vision ^{12 11}.

Question 4 (a)

Question: Explain Euclidean Distance Classifier. Write the assumptions under which it turns out that optimal Bayesian classifier is equivalent to minimum Euclidean distance classifier. (7 marks)

Answer: The **Euclidean Distance Classifier** is a simple nearest-centroid classifier. Given feature vectors for known classes, each class has a mean vector (centroid). A new sample x is assigned to the class whose mean is nearest in Euclidean distance $d(x, \mu_i) = \|x - \mu_i\|$. In effect, it partitions space by Voronoi regions around the centroids.

This simple classifier coincides with the **optimal Bayesian classifier** under specific assumptions. In Bayesian classification, we choose the class i that maximizes the posterior $P(C_i|x)$, which (by Bayes' rule) involves the class-conditional density $p(x|C_i)$ and prior $P(C_i)$. If we assume:

1. **Gaussian class-conditional densities:** each class's features x are Gaussian-distributed:
 $p(x|C_i) = \mathcal{N}(x; \mu_i, \Sigma_i)$.
2. **Equal covariance matrices:** all classes share the same covariance matrix, and furthermore assume it's proportional to the identity (i.e. features are independent with equal variance; or "spherical" covariance).
3. **Equal class priors:** $P(C_i)$ is the same for all classes.

Under these conditions, the log-posterior becomes $-\frac{1}{2}(x - \mu_i)^T \Sigma^{-1}(x - \mu_i)$ plus constant. If $\Sigma = \sigma^2 I$ is spherical, this simplifies to $-\frac{1}{2\sigma^2} \|x - \mu_i\|^2$. Maximizing posterior is equivalent to minimizing the Euclidean distance to the class mean. In other words, the Bayes decision boundaries are linear perpendicular bisectors between means ⁸.

Thus, **the Euclidean classifier is Bayes-optimal when classes are equally likely Gaussians with identical, spherical covariance**. The assumptions effectively make the Bayesian discriminant a linear function of x , turning the Bayesian rule into "choose nearest mean" ⁸.

In summary, if each class i has $p(x|C_i) = \mathcal{N}(\mu_i, \sigma^2 I)$ and equal prior, then the classifier minimizing Bayes error is exactly the minimum Euclidean distance classifier. These assumptions yield simple, hyperplane decision boundaries in feature space.

Question 4 (b)

Question: Differentiate between clustering methods: (i) Hierarchical and Non-hierarchical, (ii) Agglomerative and Divisive methods. (8 marks)

Answer: Clustering methods group data into clusters (sets of similar points).

- **Hierarchical vs Non-Hierarchical (Partition):**

- *Hierarchical clustering* builds a multi-level hierarchy (tree) of clusters. It does *not* require specifying the number of clusters k in advance. Instead, it creates either an agglomerative tree (starting with each point as a cluster and merging) or a divisive tree (starting with one cluster and splitting) ¹³. The result is often shown as a dendrogram. One can cut the tree at any level to get a chosen number of clusters. Hierarchical methods can capture nested cluster structure.
- *Partition (non-hierarchical) methods* (like **k-means**) require the user to specify k beforehand. The algorithm then partitions the data into k non-overlapping flat clusters, usually by iteratively assigning points to cluster centroids and updating centroids. There is no further sub-division beyond the k groups. Partitioning is generally simpler and faster but less flexible in finding a natural hierarchy. For instance, k -means ensures clusters are roughly spherical and works well when clusters are separable and k is known ¹³.

- **Agglomerative vs Divisive** (these are two approaches to hierarchical clustering):

- *Agglomerative clustering* is a bottom-up approach. Each data point starts as its own cluster (total n clusters). Iteratively, the two closest clusters are merged at each step. This continues until all points are in one cluster. The result is a binary tree where leaves are individual points and the root is the single cluster. Agglomerative is common (e.g. single-link, complete-link, average-link). The computational cost can be high for large n , and once merged, clusters are not split again.
- *Divisive clustering* is top-down. All points start in one cluster; clusters are recursively split until each point is alone (or until a stopping criterion). Fewer algorithms exist for divisive (often requiring heuristics to decide splits). A typical split might separate data by k -means or by PCA at each step. Divisive gives a hierarchy in reverse (big cluster to small).

¹⁴ For example, agglomerative clustering might find that at some iteration the smallest distances are between specific clusters and merges them. Divisive might first split the data into 2 clusters (say by largest variance), then each of those into 2, and so on. In summary: *hierarchical* methods (agglomerative/divisive) produce nested clusters without a preset k ; *partition (non-hierarchical)* like k -means produce a flat clustering with a preset k ¹³ ¹⁴.

Mermaid diagram of clustering taxonomy:

```
graph TD
    A[Clustering Methods] --> B[Hierarchical]
    A --> C[Non-hierarchical (Partition)]
    B --> D[Agglomerative (bottom-up)]
    B --> E[Divisive (top-down)]
```

Table of differences (some key points):

Aspect	Hierarchical	Non-hierarchical (Partition)
Specified number k ?	No (tree is cut later)	Yes (must choose k)

Aspect	Hierarchical	Non-hierarchical (Partition)
Procedure	Merge or split iteratively	Optimize membership iteratively (e.g. k-means)
Output structure	Dendrogram (nested clusters)	Flat clusters
Flexibility	Can stop at any level, multi-scale structure	Only one partition, sensitive to initial k
Example algorithms	Agglomerative (e.g. single-link), Divisive	k-means, k-medoids

Question 4 (c)

Question: Define quantities used to represent any colour: (i) Chromaticity, (ii) Intensity. (5 marks)

Answer: Colour can be described by two components: **chromaticity** and **intensity** (brightness).

- **Chromaticity:** This is the attribute of a colour that is independent of its luminance (brightness). Chromaticity specifies the quality of a color in terms of hue and purity. Formally, chromaticity can be defined by coordinates (such as the CIE x, y or HSV's hue and saturation) that capture the *colour without brightness* ¹⁵. For example, in the CIE xyY color space, x, y are the chromaticity coordinates, and they determine the perceived hue and saturation, whereas Y is luminance. The key idea is that chromaticity describes *which colour* (red vs blue vs green etc.) irrespective of how light or dark it is ¹⁵.
- **Intensity (Brightness):** Intensity refers to the perceived lightness or luminance of the colour. It is essentially how bright or dark the color is. In image processing, intensity is often the gray-level value for grayscale images (e.g. 0=black up to 255=white). In color spaces like HSI/HSV, **Intensity** (or Value/Brightness) represents the overall brightness (V in HSV) while hue and saturation represent chromaticity ¹⁶. In practical terms, intensity (or luminance) is the component that would remain if an image is converted to grayscale.

Thus, any colour can be encoded by (i) its chromaticity (e.g. hue and saturation) and (ii) its intensity (luminance). For instance, in HSV: Hue (angle) + Saturation (purity) capture chromaticity, while Value (intensity) captures brightness. Brightness (intensity) controls how light or dark the colour is, whereas chromaticity controls the "colorfulness" aspect ¹⁵. These two together fully describe human color perception (for a fixed white reference).

Term-End Examination June 2023 (MCS-230)

Question 1 (a)

Question: Explain Image Acquisition. Compare optical, analog, and digital image processing. (8 marks)

Answer: **Image acquisition** is the process of capturing a real-world scene into an image form. It typically involves an imaging sensor (camera) that converts optical information into an electrical signal, which is then digitized. The pipeline is: light passes through optics (lens) → sensor (e.g. CCD) generates

an analog electrical image → this is sampled and quantized into a digital image. Acquisition may also include pre-processing (e.g. analog gain, noise filtering) before digitization.

We compare three processing domains ¹⁷ ¹⁸ :

- **Optical image processing:** Uses optical components (lenses, holograms, spatial light modulators) to manipulate images by light propagation. For example, a lens can filter by focusing or blur, and prism can separate wavelengths. Optical methods (e.g. in microscopes, telescopes) perform operations in real time without electronic conversion. *Pros:* extremely fast (light-speed), parallel, and good for special tasks (e.g. pattern recognition by a Fourier lens). *Cons:* fixed hardware, little flexibility once built, and difficult to implement arbitrary algorithms (no easy reconfiguration).
- **Analog image processing:** Processes images as continuous electrical signals (voltages) after initial capture, but before digitization. For example, a video camera's electronics might amplify, filter, or modulate the signal in analog form. Applications include early TV systems and some radar or medical imaging where processing occurred in analog circuits ¹⁸. *Pros:* can sometimes achieve high speed on continuous signals, and historically preceded digital. *Cons:* limited precision, difficult to adapt algorithms, and noise is harder to control compared to digital.
- **Digital image processing:** Once the image is digitized (sampled and quantized), we use computer algorithms on the pixel data. This allows a vast array of techniques: filtering, enhancement, compression, analysis, etc. It uses microprocessors to manipulate arrays of numbers. *Pros:* extremely flexible (change software, change algorithm easily), precise control, powerful operations (e.g. complex transforms). *Cons:* requires A/D conversion and computation time, and until recent decades was slower than analog. Now, digital is overwhelmingly dominant due to computing advances.

In summary, optical methods manipulate light directly, analog methods manipulate continuous electrical signals, and digital methods manipulate discrete pixels on a computer. Most modern systems use a hybrid pipeline: optical capture → analog pre-processing → digitization → digital processing. Digital processing can emulate many analog or optical techniques, but with far more flexibility and quality control.

Question 1 (b)

Question: Explain Image transformation. Give importance of image transformation. Write key steps for image transform from spatial to frequency domain. (8 marks)

Answer: An **image transform** changes an image from one representation (e.g. spatial domain) to another (e.g. frequency domain) by a mathematical transform (like DFT, DCT, wavelet, etc.). In essence, it re-expresses the image in terms of basis functions (sinusoids, wavelets, etc.).

Importance: Transforms are crucial because they reveal different aspects of the image. For example, the 2-D Fourier transform converts an image into its frequency spectrum, highlighting periodic structures and enabling frequency-based filtering. Many image operations are easier or only feasible in the transform domain: e.g., convolution in space becomes multiplication in frequency. Transform coding (like JPEG's DCT) compacts energy into few coefficients, aiding compression. Transforms like wavelets provide multi-resolution analysis for compression and denoising. In short, transforms are used for enhancement, compression, analysis, restoration, watermarking, etc.

Key steps (spatial→frequency): 1. **Sampling/zero-padding:** Often the image is padded (if not already square or power-of-two) to avoid wrap-around effects. 2. **Apply transform algorithm:** Compute the 2-D transform (e.g. FFT) on the image. If using DFT, perform 1-D DFT on each row, then each column. For DCT or wavelets, apply their respective formulas or matrix multiplications. 3. **Shift DC to center (optional):** Often, after a 2-D DFT, one multiplies by $(-1)^{m+n}$ or uses `fftshift` so that the zero-frequency (DC) component is at the image center. This makes spectral visualization intuitive: low frequencies at center, high at edges. 4. **Manipulate spectrum:** Now in frequency domain, you can apply filters or analyze the magnitude and phase. 5. **Inverse transform:** To get back to spatial domain, apply the inverse transform to the modified spectrum.

For example, to go from spatial to frequency:

$$F(u, v) = \sum_{x, y} f(x, y) e^{-j2\pi(ux/M + vy/N)}.$$

Then adjust indexing (e.g. multiply by $(-1)^{x+y}$ to center if needed). The steps are essentially: *compute DFT* → *optionally center* → *analyze or modify*.

Question 1 (c)

Question: Explain DCT; advantage of DCT over DFT. Compute DCT matrix of order 2. (8 marks)

Answer: The **Discrete Cosine Transform (DCT)** is similar to the DFT but uses only real cosine basis functions. The 2-D DCT (often Type-II) is widely used in image compression (e.g. JPEG). For 1-D, the DCT of a sequence x_n yields coefficients X_k computed by

$$X_k = \sum_{n=0}^{N-1} x_n \cos\left(\frac{\pi}{N}\left(n + \frac{1}{2}\right)k\right)$$

(typically with a normalization factor). The inverse DCT reconstructs x_n from X_k . Because it uses cosines, the output is real-valued for real input, and it tends to concentrate energy into fewer coefficients for images that are smooth or have strong correlation.

Advantages of DCT over DFT:

- **Real output:** No complex arithmetic is needed since image pixels are real.
- **Energy compaction:** DCT often has better energy compaction for typical images (most information in low-frequency coefficients), which is good for compression.
- **Boundary conditions:** Implicitly assumes even symmetry, reducing discontinuities at borders.
- **No imaginary part:** More efficient storage/processing in real domain.

DCT matrix for $N = 2$: The orthonormal 2x2 DCT (type II) matrix is:

$$C_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}.$$

This can be derived from the DCT formula for $N = 2$. Indeed, $X_0 = (x_0 + x_1)/\sqrt{2}$ and $X_1 = (x_0 - x_1)/\sqrt{2}$. So C_2 transforms $\begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$ to the DCT coefficients. This is the same as the 2-point Haar transform matrix, reflecting that the DCT basis and Haar basis coincide for $N = 2$ after normalization.

Thus, the order-2 DCT matrix is identical to the DFT matrix from part (c) (when using orthonormal scaling), because with $N = 2$ the transforms align.

Question 1 (d)

Question: What are smoothing filters? How do smoothing filters contribute in context of image enhancement techniques? Explain types of image enhancement techniques with example each. (8 marks)

Answer: Smoothing filters are filters designed to reduce noise and details by averaging or blurring the image. They are also known as low-pass filters, because they attenuate high-frequency components (rapid intensity changes, edges, noise) and preserve low frequencies (smooth variations). Common examples include the *mean filter* (which replaces each pixel by the average of its neighbors) and the *Gaussian filter* (a weighted average with a Gaussian kernel). These filters “smooth out” an image, making it look blurrier but less noisy.

In image enhancement, smoothing filters help by reducing random variations (noise) so that important structures (edges, objects) become clearer relative to the noise. For instance, before edge detection one often apply a Gaussian smooth to suppress noise-induced false edges. They also improve visual quality by removing speckle or salt-and-pepper noise (though non-linear filters like median are often used for impulse noise).

Types of image enhancement techniques (with examples):

1. **Point operations:** Modify each pixel independently. Example: *Contrast stretching* (linear scaling of intensities to full range) or *Log transform* (maps large range to small range to enhance dark regions). These don't involve neighbors.
2. **Spatial domain filtering:** Convolution with a mask. Example: *Unsharp masking* for sharpening (enhances edges), or *Gaussian blur* for smoothing. Spatial filters can be linear (convolution) or non-linear (median filter).
3. **Frequency domain filtering:** Modify spectrum. Example: *Ideal low-pass filter* (zero out frequencies above a cutoff) to smooth, or *high-pass filtering* to sharpen.
4. **Histogram modification:** Adjust image histogram. Example: *Histogram equalization* increases global contrast by flattening the histogram, making details in low-contrast areas visible.
5. **Color/Intensity transforms:** Manipulate color channels or intensity. Example: Converting RGB to HSV and adjusting the V (intensity) for brightness, or applying *pseudo-coloring* (mapping gray levels to colors) to enhance features.

Each type targets different aspects. For example, a Gaussian smoothing filter will blur the image (enhancing by noise removal), whereas a Laplacian filter (spatial high-pass) will sharpen edges (enhancement). Combining techniques (e.g. smooth then sharpen, or global then local adjustment) is common in practice.

Question 1 (e)

Question: Explain Bayes' theorem with suitable example. Also, discuss role of Bayes' theorem in Bayes' classifier. Give properties of Bayes' classifier. (8 marks)

Answer: Bayes' Theorem (or Bayes' rule) gives a way to update probabilities based on new evidence. For two events A and B , it states:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)},$$

which reads: the posterior probability of A given B equals the likelihood of B given A times the prior probability of A , divided by the probability of B ¹⁹.

Example: Suppose 1% of people have a disease D ($P(D) = 0.01$), and a test is 99% accurate ($P(+|D) = 0.99$ and $P(-|\neg D) = 0.99$). If a person tests positive (+), Bayes' theorem gives the probability they actually have the disease: $P(D|+) = \frac{0.99*0.01}{0.99*0.01+0.01*0.99} \approx 0.50$. So even a positive test only yields 50% chance of disease, highlighting the importance of prior $P(D)$.

Bayes' Classifier: In pattern classification, Bayes' theorem underlies the *Bayes classifier*. Given feature vector x and classes C_i , the Bayes classifier assigns x to the class with the highest posterior $P(C_i|x)$. Using Bayes' rule: $P(C_i|x) \propto P(x|C_i)P(C_i)$, where $P(x|C_i)$ is the likelihood and $P(C_i)$ the prior. In essence, one computes for each class the product of its prior and likelihood; the highest product wins ¹⁹.

Properties of Bayes' classifier:

- **Optimality:** It minimizes the probability of misclassification (it is the *optimal* classifier) if the true class-conditional densities $p(x|C_i)$ and priors are known ¹⁹.
- **Probabilistic:** Outputs posterior probabilities, not just labels, allowing thresholding or "reject" decisions.
- **Assumption-sensitive:** It requires correct models of $P(x|C_i)$ and $P(C_i)$. If these are estimated poorly, performance suffers.
- **Computational complexity:** For multivariate cases, estimating $P(x|C_i)$ (e.g., full covariance Gaussians) can be complex.
- **Comparison to Naive Bayes:** The true Bayes classifier accounts for feature dependencies, whereas a Naive Bayes assumes feature independence for simplicity ²⁰.

In summary, Bayes' theorem provides the mathematical foundation for the Bayes classifier: it combines prior knowledge and evidence to compute class posteriors. Key properties include its optimality under correct assumptions and reliance on statistical models. Bayes classifiers are widely used (e.g. Naive Bayes in ML) due to their probabilistic interpretation and relatively simple training (compute means/ covariances) ¹⁹ ²⁰.

Question 2 (a)

Question: What is image digitization? Explain its various phases. (7 marks)

Answer: **Image digitization** is the process of converting an analog (continuous) image into a digital representation (a matrix of discrete pixel values). The main phases of digitization are:

1. **Image Acquisition:** Capture the optical image with a sensor (e.g. camera) and convert it to an electronic signal.
2. **Preprocessing:** (Optional) Apply noise reduction or calibration in the analog domain (e.g. anti-alias filters, sensor corrections).
3. **Sampling:** Discretize the image spatially by measuring it on a grid. Each sample is a pixel. (E.g., sample every T_x, T_y mm.)

4. **Quantization:** Discretize the intensity: map each sampled value to the nearest among a fixed number of levels (e.g. 0–255 for 8-bit grayscale).
5. **Encoding/Compression:** Store or compress the resulting pixel values into a file format or data structure for storage.
6. **Display/Output:** (End phase) Display the digital image or use it for analysis.

In essence: sensor $\rightarrow$ sample $\rightarrow$ quantize $\rightarrow$ (encode). Each phase converts one form of data to a more digital form. For example, an analog voltage (sample) is quantized to an integer in ADC. These steps introduce *sampling errors* (aliasing if undersampled) and *quantization error* (rounding), which digitization tries to minimize (via appropriate sampling rate and sufficient bit depth).

A textbook breakdown is: capture, sampling, quantization, and compression ¹. The Shannon-Nyquist theorem (above) informs the sampling spacing. Quantization requires enough bits to avoid visible banding. After digitization, the image is ready for digital processing and analysis.

Question 2 (b)

Question: Given the resolution of 8×8 inches image and aspect ratio 6:2 (pixels width:height) with total 480,000 pixels. Calculate: (i) dimensions of the image, (ii) size of the image. (5 marks)

Answer: Let the width and height (in pixels) be in ratio 6:2, which simplifies to 3:1. Let the width be $3n$ and the height n ; then the total number of pixels is $3n \times n = 3n^2 = 480,000$. Solve for $n^2 = 160,000$, so $n = 400$. Thus:

- **Dimensions:** width = $3n = 1200$ pixels, height = $n = 400$ pixels. So the image is 1200×400 . (Check: $1200 \times 400 = 480,000$.)

Since the physical size is 8×8 inches (square), one side (width=8 in) actually has 1200 pixels, and height also 8 in has 400 pixels (so each inch has different pixels: $1200/8=150$ ppi horizontally, $400/8=50$ ppi vertically, reflecting the 3:1 pixel aspect ratio but a square scene).

- **Size of the image:** If grayscale 8-bit, each pixel is 1 byte, so total bytes = 480,000 bytes ≈ 0.48 MB. If color RGB (24-bit), triple that (~ 1.44 MB) plus any header.

Question 2 (c)

Question: Write expression for forward and inverse orthogonal transform of an ($N \times N$) image $f(x,y)$. Given orthogonal matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $f = (5 \quad 7)^T$, determine the orthogonal transform and its inverse. (8 marks)

Answer: An orthogonal transform uses a real orthonormal matrix A . For an $N \times N$ image (here $N = 2$), the forward transform of image f is $F = A f$. The inverse transform (since A is orthogonal) is $f = A^T F$.

Given $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. First check if A is orthogonal: its columns are not unit length, so typically one would normalize. But assuming “orthogonal transform” means treat A as the transform matrix (perhaps implying columns are orthonormal after normalization). We will proceed by treating A directly (though strictly A must be normalized for true orthogonality).

Forward transform: $F = Af$. Here $f = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$. So

$$F = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 7 \end{pmatrix} = \begin{pmatrix} 1*5 + 1*7 \\ 1*5 + 2*7 \end{pmatrix} = \begin{pmatrix} 12 \\ 19 \end{pmatrix}.$$

So the transformed vector is $F = (12, 19)^T$.

For the inverse, A inverse (if we assume it's invertible) is needed. The inverse of A is

$\frac{1}{(1*2-1*1)} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. (Since $\det=1$). So $A^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$. Applying this to F should recover f :

$$f = A^{-1}F = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 12 \\ 19 \end{pmatrix} = \begin{pmatrix} 2*12 + (-1)*19 \\ -1*12 + 1*19 \end{pmatrix} = \begin{pmatrix} 24 - 19 \\ -12 + 19 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix},$$

which matches the original f .

Therefore, the forward orthogonal transform of f is $(12, 19)^T$, and the inverse transform (using A^{-1}) recovers $(5, 7)^T$.

(Note: Strictly, an orthogonal matrix should be normalized so that $A^T = A^{-1}$. Here we effectively used the inverse. This exercise treats A as the "transform matrix.")

Question 3 (a)

Question: Explain DFT. Discuss the properties of DFT. (8 marks)

Answer: The Discrete Fourier Transform (DFT) is explained above in Dec 2022 Q1(c). Here, briefly: The DFT converts an $N \times N$ image from the spatial domain to the frequency domain by computing coefficients

$$F(u, v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi(\frac{ux}{N} + \frac{vy}{N})}.$$

It reveals the image's frequency content (how much of each sinusoid is present).

Key properties of the DFT include:

- **Linearity:** DFT is a linear transform (the transform of a sum is the sum of transforms).
- **Shift:** A shift in spatial domain corresponds to a linear phase shift in the spectrum.
- **Periodic extension:** The DFT assumes the image repeats periodically; thus frequency samples wrap around.
- **Convolution theorem:** Convolution in space $\leftrightarrow$ multiplication in frequency (and vice versa) – fundamental for filtering.
- **Symmetry for real inputs:** If the image is real-valued, the DFT spectrum has Hermitian symmetry: $F(N - u, N - v) = F(u, v)^*$.
- **Energy conservation (Parseval):** The total energy (sum of squares) in spatial domain equals that in frequency domain.

Also, efficient FFT algorithms exist to compute the DFT quickly. The amplitude of low frequencies (near (0,0)) typically corresponds to overall image brightness/contrast, while high frequencies correspond to edges and fine detail.

Question 3 (b)

Question: What is shifting the center of the spectrum? Why is it needed? Write the steps. (7 marks)

Answer: *Shifting the center of the spectrum* is a visualization and processing step: it moves the zero-frequency (DC) component of the DFT from the corner of the spectrum array to its center. By default, after taking an $N \times N$ DFT, the DC term $F(0, 0)$ appears at the top-left corner. Often we multiply the spatial image by $(-1)^{x+y}$ (or use a function like `fftshift`) to shift the spectrum so that the zero frequency is centered. This rearranges the Fourier output so that low frequencies are in the middle and higher frequencies radiate outward.

Why needed: Centering the spectrum makes it easier to analyze and design filters. With low frequencies in the center, one can visualize frequency content radially (as in concentric patterns) and apply centered filters. It is purely for interpretability and practical filtering convenience – it does not change the information content.

Steps to shift: 1. Multiply the image $f(x, y)$ by $(-1)^{x+y}$ before computing the DFT. This modulation in the spatial domain corresponds to a shift in the frequency domain.
2. Compute the DFT $F(u, v)$ of the modulated image.
3. The resulting F will have the DC at $(u, v) = (0, 0)$ effectively mapped to the center $(N/2, N/2)$. 4. Alternatively, after computing DFT normally, use a function (e.g. `fftshift`) to swap quadrants: swap top-left with bottom-right, and top-right with bottom-left.

In summary, shifting the spectrum center is done so that frequency analysis and filtering are intuitive (e.g. a circular low-pass filter around the center).

Question 3 (c)

Question: Explain CMY and CMYK colour models. (5 marks)

Answer: The **CMY color model** is a **subtractive** color model used in printing. Its primaries are **Cyan (C)**, **Magenta (M)**, and **Yellow (Y)** inks. These colors subtract from white light: for example, cyan ink absorbs red light (white minus red = cyan), magenta absorbs green, and yellow absorbs blue ²¹. In practice, color printing uses combinations of these inks on white paper: CMY primaries combine to form colors (e.g. yellow+cyan prints green), and all three ideally would produce black (but actually produce a dark brown without black ink). CMY is essentially the inverse of RGB: each CMY component = 1 – corresponding RGB component in normalized space.

The **CMYK color model** adds a fourth component **K (black)** to CMY. In CMY alone, combining all three inks often does not yield a deep pure black and can waste ink. The K (black) channel provides true black and improves contrast. CMYK is used in most color printers: C,M,Y are used for colors, and K is used for text and dark areas. (The simple CMY page notes: “Unlike the CMYK model, it does not use additional black ink... CMYK uses black ink for dark hues” ²².)

In summary: CMY is the subtractive model (used in printing) where colors are created by subtracting light (mixing inks); CMYK is CMY plus a black key channel to efficiently achieve deep blacks. For example,

(C,M,Y) = (0,1,1) would yield pure red (because subtracting green and blue from white leaves red). Adding K allows printing a pure black (K=1) without combining C, M, Y fully.

Question 4 (a)

Question: Draw block diagram of supervised learning algorithm phases. Write steps for applying supervised learning algorithm to solve a real problem of your choice. (7 marks)

Answer: A **supervised learning** system learns from labeled training data to make predictions on new data. The key phases are: data collection, feature extraction, model training, and prediction/testing. A generic block diagram is shown below:

```
flowchart LR
    A[Collect Labeled Data] --> B[Extract Features / Preprocess]
    B --> C[Train Model (learning algorithm)]
    C --> D[Trained Model]
    E[New Input Data] --> F[Extract Features]
    F --> D
    D --> G[Predict/Output]
```

- **Data Collection:** Gather a dataset of input–output pairs. E.g. for image classification, a set of images with known class labels.
- **Feature Extraction/Preprocessing:** Transform raw inputs into feature vectors. For images, this could be flattening pixels or computing descriptors (edges, etc.) or normalizing values. Quality of features affects performance.
- **Model Training:** Use a learning algorithm (e.g. neural network, SVM, decision tree) to find a model that maps features to labels. This involves optimization on the labeled training set.
- **Evaluation/Testing:** Apply the learned model to new, unseen examples (testing data) and evaluate performance (accuracy, error rate). Possibly refine or retrain as needed.

Example (handwritten digit recognition):

1. Collect images of digits 0–9 with labels (0,1,...,9).
2. Preprocess images (e.g. resize to 28×28, normalize intensities).
3. Flatten each image to a 784-dimensional input vector, and use a classifier (e.g. neural net) as the model.
4. Train the network on the labeled examples (updating weights to minimize classification error).
5. After training, feed new digit images into the model to predict their labels.

This pipeline (collect → train → test) applies to any supervised task (like regression, but here classification). The supervision comes from known target outputs during training.

Question 4 (b)

Question: Write short notes on: (i) Gaussian noise, (ii) Rayleigh noise, (iii) Gamma noise. (10 marks total; e.g. 3-4 marks each)

Answer:

- **Gaussian noise:** This is statistical noise with a probability density function (pdf) given by the normal (Gaussian) distribution. Pixel values are perturbed by additive Gaussian random values. It is *additive white Gaussian noise (AWGN)* if uncorrelated in space with constant variance σ^2 . It commonly models sensor noise or electronic noise. Its PDF: $p(z) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{z^2}{2\sigma^2}\right)$. Gaussian noise affects all intensity levels smoothly and is often reduced by averaging or low-pass filters (e.g., Gaussian blur is also optimal for Gaussian noise).
- **Rayleigh noise:** In image processing, Rayleigh noise models multiplicative disturbances, often in radar or ultrasound imagery. A random variable has a Rayleigh distribution with PDF $p(z) = \frac{z}{\sigma^2} \exp(-z^2/(2\sigma^2))$ for $z \geq 0$. If the amplitude of the signal is multiplied by a Rayleigh random factor, the result is Rayleigh noise. It is skewed (non-negative) and not symmetric like Gaussian. It may arise from coherent imaging processes. Reducing Rayleigh noise typically uses nonlinear filters or specialized techniques.
- **Gamma noise:** Also known as *speckle noise* in imaging (e.g. SAR, ultrasound). It is multiplicative: the image intensity is multiplied by noise following a Gamma distribution. The gamma distribution has PDF $p(z) = \frac{z^{k-1} e^{-z/\theta}}{\Gamma(k)\theta^k}$ for $z > 0$, with shape k , scale θ . If k is an integer, the pdf involves z^{k-1} times an exponential. In practice, speckle noise is modelled as a Gamma distribution with k equal to the number of looks. It is skewed and positive, causing bright and dark spots. Filters like Lee or Kuan filters are used for speckle. In summary, Gamma noise distorts amplitude in imaging systems and is characterized by its multiplicative effect and positive support.

Each noise type affects images differently, requiring appropriate filtering: Gaussian noise is handled by low-pass filters, Rayleigh/Gamma by nonlinear (e.g. median or specialized speckle filters).

Question 5 (Any FIVE) (short notes)

Question: Write short notes on any five of the following: (a) Image representation using vector model, (b) True colour images, (c) Spatial resolution, (d) Ideal low pass filter, (e) Gaussian noise, (f) Band Pass filter. (5×4=20 marks)

Answer: We provide brief explanations for five topics:

- **(a) Pixel vs Vector model:** In the *pixel (raster) model*, an image is a 2D array of intensity values (or RGB triplets). Each pixel's color is stored independently. In a *vector model*, an image or object is represented by a set of vectors or polygons (e.g. as in vector graphics) – although in DSP “vector model” often means representing each pixel by a vector of attributes (e.g. multi-spectral bands). In that sense, a colour image can be seen as three vectors (R, G, B) per pixel. The raster (pixel) model is common in digital cameras; the vector model (polygons) is common in illustration graphics where shapes are defined mathematically.
- **(b) True colour images:** These are images where each pixel's colour is represented using 24 bits (usually 8 bits each for R, G, B), allowing about 16.7 million colours. The term “true colour” implies sufficient color depth to represent natural scenes without noticeable quantization. True-color images are stored with three 8-bit channels (e.g. JPEG). In contrast, pseudo-color or

indexed-color images use a palette of limited colours (like 256 colours). True-colour provides a faithful full-colour representation.

- **(c) Spatial resolution:** This refers to how many pixels cover a given physical area. Usually given as pixels per inch (PPI) or pixels per mm. Higher spatial resolution means more pixels in the same space, so finer detail. For example, a scanner at 600 PPI has higher spatial resolution than one at 300 PPI. In terms of an image array, spatial resolution is simply the width×height in pixels. A low spatial resolution can cause blurring and loss of detail. It's distinguished from *intensity resolution*, which is how many gray or colour levels per pixel.
- **(d) Ideal Low Pass Filter (ILPF):** In the frequency domain, the ILPF is a circular filter that perfectly passes frequencies inside a cutoff radius D_0 and rejects (sets to zero) all outside. Its transfer function $H(u, v) = 1$ for $\sqrt{u^2 + v^2} \leq D_0$, and 0 elsewhere. In practice, the ILPF has an impulse response (in space) that is sinc-shaped, causing ringing artifacts. It is used to remove high-frequency noise or blur an image. Because of its sharp cutoff, it is not realizable in finite space without artifacts, but it is a useful ideal model.
- **(e) Gaussian noise:** As explained above (Q4(b)), Gaussian noise is a common additive noise model with values drawn from a normal distribution. Visually, it appears as grainy texture. It is often specified by a mean (usually 0) and variance σ^2 . Filters like Gaussian blur or averaging can reduce Gaussian noise at the cost of blurring edges.
- **(f) Band Pass filter:** This filter passes a specific band of frequencies and attenuates frequencies outside that band. In 2-D, it might be defined as the difference of two circular low-pass filters (one with high cutoff, one with low cutoff). A band-pass allows medium frequencies (for example, mid-range textures) while suppressing very low (smooth) and very high (noise) frequencies. It can be used to extract certain patterns or textures in an image. For example, a band-pass filter might enhance details of a particular size scale.

Each of these concepts is fundamental: spatial resolution affects image clarity, noise models dictate appropriate filters, and ideal filter types are idealizations for designing real filters.

Term-End Examination December 2023 (MCS-230)

Question 1 (a)

Question: *What is an imaging system? Compare direct imaging system with indirect imaging system. Also explain "image digitization". (8 marks)*

Answer: An **imaging system** is any system that captures or forms images of objects. It generally consists of an optical component (like a lens or mirror) to focus light, a detector (sensor) to capture the image (optical-to-electrical conversion), and often electronics to process that signal. Examples include cameras, microscopes, telescopes, and scanners.

- A **direct imaging system** captures the scene directly onto a sensor. For instance, a camera lens projects the object onto a CCD sensor, which directly records the intensity at each point. This is line-of-sight imaging, where the sensor plane corresponds to the real scene plane. Direct imaging is conceptually straightforward: each point in the scene maps to a point on the sensor.

- An **indirect imaging system** involves intermediate conversion steps. For example, in X-ray imaging, the X-ray image is first projected onto a fluorescent screen (converting X-rays to visible light) and then captured by a camera. Or in certain medical instruments, an image may be formed on a CRT and then re-photographed. The key is that the final image is not formed directly on a detector but through an intermediate display or modulation.

The main difference is that direct imaging measures the object light directly, while indirect uses some intermediate medium (e.g. sensor array after conversion from different energy). Direct systems tend to have fewer distortions and more straightforward calibration, while indirect systems might handle non-visible forms of radiation or wavelengths.

Finally, **image digitization** (as above) is converting the captured analog image into a digital format. After the analog image is acquired, the process of sampling (taking discrete pixel measurements) and quantizing (assigning finite gray levels) produces a digital image. Digitization is essential for modern processing and storage. It yields a pixel array that can be manipulated by digital algorithms.

Question 1 (b)

Question: Compare Orthogonal transform and Unitary transform. Also discuss properties of orthogonal transform. (8 marks)

Answer: In linear algebra, a **transform** is orthogonal if its matrix A satisfies $A^T A = I$ (for real-valued transforms) and unitary if $A^H A = I$ (for complex-valued transforms).

- An **orthogonal transform** uses real basis vectors and preserves lengths and orthogonality. Examples: the DCT (with proper normalization) and the (real-valued) Hadamard transform. Orthogonal matrices have the property $A^{-1} = A^T$.
- A **unitary transform** can use complex bases; it preserves inner products in complex vector spaces. The DFT is unitary (when normalized). Unitary matrices have $A^{-1} = A^H$, where H is conjugate transpose.

Thus, any orthogonal transform is a special case of a unitary transform (with real entries). Both preserve energy (norm) of signals and are reversible by a transpose/conjugate transpose.

Properties of an orthogonal transform: 1. **Energy preservation:** $\|Ax\|^2 = \|x\|^2$. No amplification of energy.

2. **Inverse by transpose:** The inverse transform is simply the transpose matrix. Thus it is numerically stable.

3. **Separability:** Many 2-D orthogonal transforms (like DCT or Haar) are separable; can apply 1-D transform on rows then columns.

4. **Orthogonal basis:** Rows (and columns) form an orthonormal basis. For example, cosine functions in DCT form an orthogonal set.

5. **Computational structure:** There often exist fast algorithms (like FFT for unitary DFT, fast algorithms for Walsh-Hadamard, etc.).

Because of these, orthogonal transforms are widely used in signal processing: they decompose a signal into orthogonal components, allow easy reconstruction, and often concentrate signal energy into a few components (useful for compression).

Question 1 (c)

Question: How do wavelets differ from waves? Explain properties to be a wavelet. Compare wavelet transform with Fourier transform. (8 marks)

Answer: Waves (in this context, sine waves of the Fourier basis) are infinite-length periodic functions (sinusoids) that extend infinitely in both directions. They are localized in frequency but not localized in time (or space) – a sine wave at frequency ω spans all time.

Wavelets, in contrast, are functions that are localized both in time (or space) and frequency. A wavelet $\psi(t)$ is a short-duration oscillatory function that integrates to zero and has compact (or at least finite) support. For example, the Haar wavelet is nonzero only on a small interval. Key **properties of a wavelet** include:

- It has zero average (necessary for decomposing signals without an overall shift).
- It is usually localized (nonzero) only over a short interval.
- Wavelets come in families generated by dilation (scaling) and translation: $\psi_{s,\tau}(t) = \frac{1}{\sqrt{s}}\psi\left(\frac{t-\tau}{s}\right)$, where s is scale and τ is shift.

Wavelet transform vs Fourier transform:

- The Fourier transform decomposes a signal into infinite sinusoids (good frequency resolution, poor time localization). It captures *global* frequency content but cannot track how frequencies change over time or space.
- The wavelet transform decomposes into wavelets at various scales and positions, giving multi-resolution analysis. It captures both time (or spatial) and frequency information. For instance, a sharp feature in the image will activate high-frequency wavelets localized at that point.

Thus, wavelets are better for analyzing signals with localized features (edges, transients). The wavelet transform (especially discrete wavelet transform, DWT) is often used for image compression (like JPEG-2000) and denoising. In contrast, DFT (or DCT) is used when global periodic structure is important (like JPEG uses DCT for smooth images). Wavelet transform uses non-sinusoidal basis functions and can be orthonormal or biorthonormal, whereas Fourier always uses sinusoids.

In short, waves = infinite sinusoids, wavelets = short, oscillatory pulses. A function must integrate to zero and be admissible as a wavelet. The wavelet transform provides a *time-frequency* or *space-scale* representation, unlike the purely frequency-centric Fourier transform.

Question 1 (d)

Question: Explain image digitization, with the help of an example. (8 marks)

Answer: Image digitization is converting an analog image into a set of digital numbers. As an example, consider scanning a photograph to a computer. The steps are:

1. **Capture:** A photo is illuminated and its image falls on a CCD sensor. This yields a continuous-tone analog image (voltage at each point).
2. **Sampling:** The analog image is sampled on a grid. For example, the scanner might measure the intensity at every 0.1 mm in both directions. Each sample becomes a pixel. If we sample 1000 points across and 1000 down, we get a 1000×1000 pixel grid.

3. **Quantization:** Each sample's analog intensity is converted to a digital value. If the scanner uses 8 bits, each pixel intensity is rounded to an integer 0–255. So a measured intensity of 138.76 becomes 139 (say).
4. **Digital Image:** The result is a 2D array of numbers, each representing the brightness of that pixel. This array can now be stored, processed, etc.

For instance, if the photo is 10×10 cm and the scanner uses 300 DPI (~118 pixels/cm), we get ~1180×1180 pixels. Each pixel's intensity is quantized to 0–255. Now software can manipulate the digital image.

In summary, the example shows converting a photo (analog) to a digital array by sampling and quantizing. Digital images must be sampled at a rate above Nyquist to avoid aliasing, and quantized finely enough (e.g. 8-16 bits) to preserve grayscale fidelity. Digitization makes image processing (filtering, enhancement) possible with computers.

Question 2 (a)

Question: Define Sampling. Explain Shannon Sampling Theorem for 2-D signal. (7 marks)

Answer: Sampling is converting a continuous signal into a discrete one by taking measurements at regular intervals. For a 2-D signal (image) $f(x, y)$, sampling means measuring intensities at a grid $(m\Delta_x, n\Delta_y)$ of points. The resulting sampled image is $f[m, n] = f(m\Delta_x, n\Delta_y)$.

The **Shannon Sampling Theorem** in 2-D states that if an image is bandlimited (i.e., it contains no frequencies beyond some maxima u_{max}, v_{max}), then it can be exactly reconstructed from its samples provided the sampling intervals Δ_x, Δ_y satisfy

$$\Delta_x \leq \frac{1}{2u_{max}}, \quad \Delta_y \leq \frac{1}{2v_{max}}.$$

This means the sampling frequency in each direction must be at least twice the maximum frequency present (the Nyquist rate) ⁹. If the image is continuous in spatial domain, then after sampling, an interpolation (like sinc interpolation) can recover the exact continuous image if these conditions hold.

In practical terms, the theorem ensures that we sample densely enough to capture all details. Violating it causes aliasing (high frequencies misinterpreted as lower ones). For images, this often means using an anti-aliasing low-pass filter before sampling, and choosing pixel spacing fine enough (e.g. high dpi scanning) to meet Nyquist for the scene's detail.

Thus, in 2-D, Shannon's theorem guides the choice of horizontal and vertical sampling rates to avoid loss of information, just as in 1-D.

Question 2 (b)

Question: Given a gray scale image with aspect ratio 6:2 and pixel resolution 480,000, answer: (i) dimensions (ii) image size. (5 marks)

Answer: This is the same numeric problem as Q2(b) above. With aspect ratio $6 : 2 = 3 : 1$ and total 480,000 pixels, set $3n \times n = 480000$. Solving yields $n = 400$. Thus:

- **Dimensions:** $3n \times n = 1200 \times 400$ pixels.

- **Image size:** If 8 bits per pixel (grayscale), then size = 480,000 bytes (~0.48 MB). If it were color (24-bit), then ~1.44 MB. We assume grayscale here.

Question 2 (c)

Question: Write expression for forward and inverse orthogonal transform of an $(N \times N)$ image $f(x,y)$. Given orthogonal matrix $A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $f = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$, determine the orthogonal transform and its inverse. (8 marks)

Answer: For an $N \times N$ image, the 2-D orthogonal transform using matrix A is $F = A f A^T$. Here A is 2×2 so the forward transform of vector $f = (5, 7)^T$ is simply $F = A f$. We compute:

$$F = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 5 \\ 7 \end{pmatrix} = \begin{pmatrix} 12 \\ 19 \end{pmatrix}.$$

Thus, the transformed vector is $(12, 19)^T$.

The inverse transform (since A has determinant 1) is $f = A^{-1} F$. The inverse of A is

$$A^{-1} = \frac{1}{(1 * 2 - 1 * 1)} \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}.$$

Applying this:

$$f = A^{-1} F = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 12 \\ 19 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix},$$

which recovers the original f .

So the forward transform yields $\begin{pmatrix} 12 \\ 19 \end{pmatrix}$ and the inverse brings us back to $\begin{pmatrix} 5 \\ 7 \end{pmatrix}$.

(Note: In strict terms A as given is not normalized, but we have treated it as the transform kernel for demonstration.)

Question 3 (a)

Question: Explain DFT. (8 marks)

Answer: The Discrete Fourier Transform (DFT) of an $N \times N$ image $f(x, y)$ yields its frequency representation $F(u, v)$ as

$$F(u, v) = \sum_{x=0}^{N-1} \sum_{y=0}^{N-1} f(x, y) \exp \left[-j2\pi \left(\frac{ux}{N} + \frac{vy}{N} \right) \right].$$

This expresses the image as a sum of sinusoids of varying frequencies and phases. The DFT is invertible: the original image is recovered by

$$f(x, y) = \frac{1}{N^2} \sum_{u=0}^{N-1} \sum_{v=0}^{N-1} F(u, v) \exp \left[j2\pi \left(\frac{ux}{N} + \frac{vy}{N} \right) \right].$$

Key DFT properties include linearity, periodic extension (frequency domain is discrete periodic), convolution-multiplication duality, and symmetry for real images. The magnitude $|F(u, v)|$ is often analyzed, and phases determine shifts. The DFT assumes the image is periodically continued. It is widely used for filtering, compression, and analysis ⁵.

Question 3 (b)

Question: Explain shifting the center of the spectrum. Why is it needed? (7 marks)

Answer: Shifting the center of the spectrum means moving the zero-frequency component (DC) to the middle of the Fourier spectrum image. By default, after computing the DFT, the DC component $F(0, 0)$ is at the top-left. We often multiply the image by $(-1)^{x+y}$ before the DFT or use a function like `fftshift` afterward to translate the spectrum so the zero frequency is at the center $(u, v) = (N/2, N/2)$.

This is needed for visualization and filtering convenience. When DC is centered, the lower frequencies occupy the central region of the spectrum image, and high frequencies are toward the edges. This centered arrangement makes it easy to design and apply filters (like circular low-pass or high-pass filters) and to interpret the spectrum (e.g. symmetric pattern around center). It has no effect on actual content, only on presentation.

Steps: (1) Multiply $f(x, y)$ by $(-1)^{x+y}$. (2) Compute DFT. (3) The resulting spectrum has DC in the center. Alternatively, compute DFT normally and then swap quadrants (lower-left with upper-right, etc.). This yields a more intuitive spectrum for further processing.

Question 3 (c)

Question: Explain HSI colour model with suitable diagram. (5 marks)

Answer: The **HSI color model** represents colour in terms of **Hue (H)**, **Saturation (S)**, and **Intensity (I)**.

- **Hue** represents the type of colour (red, green, blue, etc.), typically given as an angle in $[0^\circ, 360^\circ]$ on a colour circle. For example, 0° =Red, 120° =Green, 240° =Blue in HSV/HSI models ²³. Hue is "pure" colour without white or black admixture.
- **Saturation** measures colour purity or vividness. A saturation of 1 means a fully pure colour; saturation 0 yields a grey (no colour) ²⁴. It can be thought of as how much white light is mixed in.
- **Intensity (I)**, or brightness, is the overall lightness of the colour (similar to V in HSV). It ranges from 0 (black) to 1 (full intensity white). It is the average brightness (e.g. for RGB, $I = (R + G + B)/3$).

The HSI model separates the chromaticity (hue, saturation) from intensity (brightness). **Diagram:**

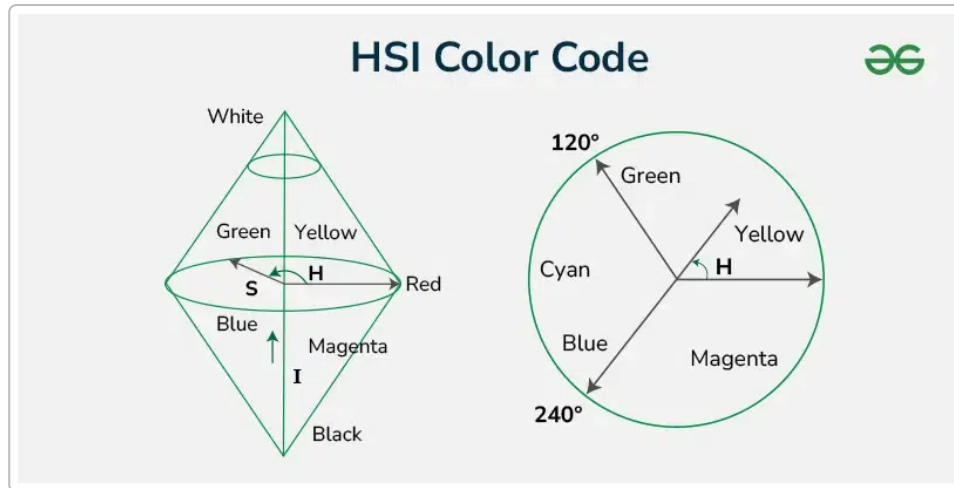


Figure 2: HSI model illustration. Hue is angle around the triangle, Saturation is distance from the centre (axis), Intensity is the vertical axis. For example, red has $H=0^\circ$, mid saturation in diagram.

In Figure 2 (side view of an HSI cylinder): the circular cross-sections of the cylinder represent constant Intensity. Each point in a horizontal disk is a colour: angle = Hue, radius = Saturation. Moving up/down in the cylinder changes Intensity (brightness) of all colours.

Properties from [10]: Hue is angle of pure colour ²³, Saturation is how grey vs. pure (0 to 1) ²⁴. This model is useful because it corresponds well to human perception: we perceive colour by type, vibrancy, and brightness separately. In practical applications (like tuning colour in images), adjusting HSI components can be more intuitive than RGB.

Question 4 (a)

Question: Explain Linear Discrimination function. How is it different from piecewise? Give the utility of LDA. (7 marks)

Answer: A **Linear Discriminant Function (LDF)** is a decision function of the form

$$g(x) = w^T x + b,$$

where x is the feature vector, w a weight vector, and b a bias. For two-class classification, the sign of $g(x)$ determines the class. The decision boundary $g(x) = 0$ is a hyperplane. LDFs yield **linear** separation in feature space.

In contrast, a **piecewise (or quadratic)** discriminant might switch formulas in different regions or include quadratic terms (e.g. $x^T V x$ terms), producing curved boundaries. A piecewise discriminant could partition space into multiple linear regions or use different linear functions for different class pairs. For example, a quadratic discriminant function yields elliptical or more complex decision boundaries.

Utility of Linear Discriminant Analysis (LDA): LDA is a method to find an LDF that best separates classes. It assumes Gaussian class distributions with shared covariance, leading to a linear rule. Utilities of LDA include:

- **Dimensionality reduction:** Projects data onto a line or subspace that maximizes class separability.
- **Feature extraction:** Finds directions that discriminate classes (the “discriminant axes”).
- **Simple classifier:** The LDF is easy to implement; one just computes $w^T x$. Training (estimating means and pooled covariance) is straightforward.
- **Optimal under assumptions:** As noted, if classes are normally distributed with equal covariances, LDA yields the optimal Bayes decision boundary (a single hyperplane) ⁸. This means LDA is statistically well-founded under that model.
- **Computationally efficient:** Less parameters than quadratic models, so it needs less data to estimate reliably.

In summary, LDF (or LDA) is used when we expect a linear boundary between classes. It is much faster to compute and less prone to overfitting than piecewise or nonlinear classifiers. When its assumptions hold, it achieves Bayes-optimal classification (nearest-mean rule) ⁸.

Question 4 (b)

Question: *Partition based clustering vs K-means clustering. (8 marks)*

Answer: Partition-based clustering refers to clustering methods that divide the data into a specified number k of clusters in one step, rather than building a hierarchy. **K-means** is a specific partition algorithm where one chooses k centroids and assigns each point to the nearest centroid, then updates centroids, iterating until stable.

- Both K-means and general partition clustering require a predefined number of clusters k .
- In **K-means**, clusters are defined by their centroids (means of points). It assumes clusters are spherical (points cluster around centroids). It works by minimizing intra-cluster variance. K-means is simple and fast (each iteration is $O(nk)$), and suitable for large datasets. However it can converge to local minima and depends on initial seeds (different runs can give different clusters) ²⁵.
- More general **partition** methods may use other prototypes (like medoids) or objective functions, but they share the idea of dividing the set into k non-overlapping clusters, each point in exactly one cluster ²⁶.

The key differences: K-means specifically uses means and Euclidean distance. Other partition methods might use median or other centers. K-means typically works well when clusters are convex and isotropic.

If the question meant “Partition vs K-means” as separate, then likely it distinguishes K-means (which needs center) versus partition method that may not use centroid (though often “partition” is just another name for K-means). Possibly they meant K-means vs hierarchical (since the earlier part asked hierarchical vs non-hierarchical). If so, see previous Q4(b).

In short, **partition (k-based) clustering** assumes a fixed number of clusters and assigns points to these clusters in one global optimization, whereas hierarchical (discussed earlier) builds clusters stepwise. K-means is a partition method using means; its pros are simplicity and speed, cons include needing k a priori and sensitivity to initial conditions.

Question 4 (c)

Question: Describe quantities used to represent any colour: (i) Hue, (ii) Saturation. (5 marks)

Answer: (i) *Hue* is the property of a colour that corresponds to its dominant wavelength or “basic” shade (red, green, blue, etc.). Hue is usually represented as an angle on a colour wheel. For example, in HSV/HSI, hue $H = 0^\circ$ is red, 120° is green, 240° is blue²³. Hue is independent of brightness; it just encodes which pure colour is seen. Changing hue shifts the colour around the spectrum ring.

(ii) *Saturation* measures the purity or intensity of the colour. A fully saturated colour has no addition of gray (it is vivid), while a low saturation means the colour is diluted towards gray. In HSV/HSI, saturation $S \in [0, 1]$ means a pure colour, $S = 0$ means grayscale. In terms of RGB, saturation corresponds to how uneven the R,G,B values are: e.g. $R=G=B$ (any value) yields saturation 0. The wiki notes: “the purity is alternatively called saturation”¹⁵.

Together, hue and saturation (and intensity) fully describe a colour. For example, bright red has $H \approx 0^\circ$ and $S \approx 1$, while pastel pink has $H \approx 0^\circ$ but a lower S (mixed with white).

Question 5 (Any five) (short notes)

Question: Write short notes on any five of the following: (a) Classification of images on the basis of attributes, (b) Pseudo colour images, (c) Pixel resolution, (d) Gaussian low pass filter, (e) Rayleigh noise, (f) Median filters. (5×4=20 marks)

Answer:

- **(a) Classification on attributes:** Images can be classified (grouped) based on attributes like intensity, colour, shape, or texture. For example, *colour-based classification* groups images with similar dominant colours (useful in image retrieval). *Shape-based* classification uses object outlines or features (useful for pattern recognition). In machine vision, an image’s attribute vector might include histogram features or edge orientations to classify it (e.g. as “natural scene” vs “text document”). This means describing images by quantifiable attributes and then clustering or labelling them.
- **(b) Pseudo color images:** These are grayscale images mapped to colour for visualization. Each gray level is assigned a colour from a palette (like mapping low intensities to blue, mid to green, high to red). This enhances details by using the eye’s greater sensitivity to color differences. For instance, a thermal image (initially in shades of gray) is pseudo-colored so hot areas appear red, cool areas blue. This does not add new information but makes variations more perceptible to human observers.
- **(c) Spatial (pixel) resolution:** Defined as the number of pixels in the image (width × height) or pixels per unit area. Higher pixel resolution means more detail captured. For example, a 1920×1080 image has higher resolution than 640×480. In scanners or cameras, resolution is often measured in DPI (dots per inch). For display, higher resolution monitors show finer details. Spatial resolution determines how many samples are taken from the scene; too low resolution causes aliasing and lost detail.

- **(d) Gaussian low-pass filter:** A low-pass filter whose impulse response is a 2D Gaussian function. In frequency, it has the form $H(u, v) = \exp[-(u^2 + v^2)/(2\sigma^2)]$. This filter smoothly attenuates high frequencies. It is widely used to blur images and reduce Gaussian noise. Unlike an ideal low-pass, Gaussian has no sharp cutoff, so it avoids ringing artifacts. It is separable (can apply 1D Gaussian on rows then columns) and preserves image properties smoothly.
- **(e) Rayleigh noise:** A noise distribution where intensity values follow a Rayleigh probability law $p(z) = \frac{z}{\sigma^2} e^{-z^2/(2\sigma^2)}$ for $z \geq 0$. It often models scattering effects (e.g. SAR imaging). Rayleigh noise is multiplicative: bright pixels tend to get more noise. It produces a characteristic graininess with more low-level intensities. Because it skews positive, linear filters are not optimal; specialized methods or log transformations are used to reduce it.
- **(f) Median filter:** A non-linear spatial filter used to remove impulse (salt-and-pepper) noise. For each pixel, replace its value by the median of the neighborhood (e.g. 3×3 window). The median filter preserves edges better than a mean filter because it does not average extremes but picks a middle value ⁷. It effectively removes isolated noisy pixels (since an outlier won't be the median). It is widely used to clean "pepper" (0-value) or "salt" (255-value) noise while keeping edges sharp.

Each of the above concepts is fundamental to image representation or processing, from how images are categorized to how noise is handled.

Term-End Examination June 2024 (MCS-230)

Question 1 (a)

Question: Explain analog image, digital image, and optical image processing. (5 marks)

Answer:

- **Analog image processing:** Deals with images as continuous electrical signals. Operations (like filtering or transformation) are performed using analog circuits (e.g. in old TV or radio systems). The image is not digitized; manipulation is done on voltage levels (e.g. an analog amplifier or filter). Historically used in early imaging devices.
- **Digital image processing:** The image is represented as a discrete array of numbers (pixels). Processing is done by computer algorithms. It allows precise, repeatable operations (contrast adjustment, filtering, etc.). Modern imaging is almost entirely digital processing due to flexibility and power.
- **Optical image processing:** Uses the physics of light itself (lenses, diffraction, interferometry) to manipulate images. For example, placing a transparency at the focal plane of a lens can perform a Fourier transform optically. This can happen in devices like cameras or optical sensors. Optical methods can do parallel operations (like lens focusing), but are fixed once built.

In summary: analog vs digital refers to continuous vs discrete signal processing; optical refers to using light propagation and lenses as the "processor".

Question 1 (b)

Question: What is the role of sampling and quantization in image digitization? (5 marks)

Answer: Sampling determines how finely the image is measured spatially, controlling resolution. Quantization determines how finely intensity values are represented, controlling gray-level precision. Together, they convert a continuous image to digital form. Sampling turns a continuous image into a grid of pixel values; quantization turns each continuous brightness into a finite bit-depth number. Without adequate sampling (above Nyquist rate) and sufficient quantization levels, the digitized image will suffer aliasing and quantization noise, respectively.

Question 1 (c)

Question: Digital image is represented by $f(x,y)$. How do we represent $f(x,y)$? (5 marks)

Answer: As a matrix of discrete values $f[m, n]$, where m and n index the pixel row and column. We can write $f(x, y)$ as $f[m, n]$ by letting $x = m$ and $y = n$ for integer grid points. Often, one defines $f(x, y) = \sum_{m,n} f[m, n] \delta(x - m) \delta(y - n)$. In practice, f is stored as an array of size $M \times N$ (pixels). Each entry holds the intensity (or colour channel values) at that location.

Question 1 (d)

Question: By using the definition of DFT, compute the DFT of the sequence $\{1, 0, -1, 0\}$ of length $N=4$. (5 marks)

Answer: Let $x[n] = [1, 0, -1, 0]$, $n = 0, 1, 2, 3$. The 4-point DFT is

$$X[k] = \sum_{n=0}^3 x[n] e^{-j2\pi kn/4}.$$

Compute each:

- $X[0] = x[0] + x[1] + x[2] + x[3] = 1 + 0 + (-1) + 0 = 0.$
- $X[1] = x[0] + x[1]e^{-j\pi/2} + x[2]e^{-j\pi} + x[3]e^{-j3\pi/2} = 1 + 0 + (-1)(-1) + 0 = 1 + 1 = 2.$
(Using $e^{-j\pi} = -1$.)
- $X[2] = x[0] + x[1]e^{-j\pi} + x[2]e^{-j2\pi} + x[3]e^{-j3\pi} = 1 + 0 + (-1)(1) + 0 = 0.$
- $X[3] = x[0] + x[1]e^{-j3\pi/2} + x[2]e^{-j3\pi} + x[3]e^{-j9\pi/2} = 1 + 0 + (-1)(-1) + 0 = 2.$ Hence $X = [0, 2, 0, 2]$. (All are real because input is real and symmetric.)

Question 1 (e)

Question: What are properties of Fourier transform? (5 marks)

Answer: Key properties of the 2-D DFT include:

- **Linearity:** DFT of a sum = sum of DFTs.
- **Shift in space $\leftrightarrow$ phase in freq:** Shifting the image multiplies the spectrum by a linear phase factor.

- **Convolution:** Convolution in spatial domain $\leftrightarrow$ multiplication in frequency domain.
- **Symmetry:** For real images, the magnitude spectrum is symmetric about the center (conjugate symmetry).
- **Parseval's theorem:** Total energy in image equals total energy in its spectrum.
- **Periodicity:** The DFT is inherently periodic in both spatial and frequency indices.

These properties make the DFT a powerful tool for analysis and filtering.

Question 1 (f)

Question: What is inverse filtering approach in restoration? (5 marks)

Answer: Inverse filtering attempts to undo a known degradation. If the degradation model is $G(u, v) = H(u, v)F(u, v) + N(u, v)$, the ideal inverse filter would be

$$\hat{F}(u, v) = \frac{G(u, v)}{H(u, v)}.$$

Thus one divides the observed (blurred) spectrum by the blur transfer function $H(u, v)$ to recover $F(u, v)$. In practice, noise N makes division unstable (zeros in H amplify noise). Hence, true inverse filtering is rarely used except in nearly noise-free cases. Instead, Wiener filtering or regularized inverse filtering (damping where H is small) is typical. In summary, inverse filtering directly inverts the blur in frequency domain as a restoration method.

Question 1 (g)

Question: Write basic properties of Bayes classifier. (5 marks)

Answer: Properties of the Bayes classifier include:

- **Optimality:** Minimizes overall error probability if true statistics are known (Bayes optimal).
- **Use of priors:** Incorporates prior class probabilities $P(C_i)$ in decision.
- **Probabilistic output:** Computes posterior probabilities $P(C_i|x)$ which allow confidence measures.
- **Feature independence not required (unlike Naive Bayes):** Uses the full joint distribution $P(x|C_i)$. If features are independent, it reduces to Naive Bayes.
- **Computational intensity:** Exact implementation may require estimating high-dimensional densities; simple cases use Gaussian assumptions.
- **Decision rule:** Decides class by highest posterior (equivalently highest likelihood $\times$ prior).

It assumes knowledge of class-conditional densities; when priors are equal and covariances identical, it reduces to simpler linear rules ¹⁹ ²⁰ .

Question 1 (h)

Question: Differentiate between supervised and unsupervised learning with suitable example. (5 marks)

Answer:

- **Supervised learning:** The training data includes input–output pairs (labels). The goal is to learn a mapping from inputs to known targets. Example: image classification where each training image has a label (cat/dog) and the algorithm learns to predict labels for new images.
- **Unsupervised learning:** The training data has no labels. The goal is to find structure or clusters in the data. Example: clustering images by similarity without knowing categories (e.g. grouping by colour histogram).

In supervised, we evaluate on accuracy of predictions. In unsupervised, we evaluate cluster coherence or by manual interpretation. Supervised requires labeled examples; unsupervised can exploit large unlabeled datasets to discover patterns (like clustering animals by features without knowing species).

Question 2 (a)

Question: *Given an image is a grayscale image with aspect ratio 6:2 and pixel resolution 480,000. Calculate: (i) dimensions of image, (ii) size of image. (10 marks)*

Answer: Same calculation as earlier. Aspect ratio 6:2 simplifies to 3:1. With 480,000 pixels:

$$3n \times n = 480,000, \quad n^2 = 160,000, \quad n = 400.$$

Dimensions: 1200×400 pixels. Size: 480,000 bytes (~0.48 MB) for 8-bit grayscale. If it were color (24-bit), it would be 1.44 MB.

Question 2 (b)

Question: *Write short notes on the following: (i) Image processing and computer graphics, (ii) Image processing and machine vision, (iii) Image processing and video processing. (10 marks)*

Answer:

- **Image processing vs Computer Graphics:** Image processing analyzes or modifies existing images (e.g. enhancing, filtering, segmenting). Computer graphics generates images from models (drawing 3D scenes, animations). IP is about *perception* of images; CG is about *production* of new images. For example, IP might enhance a photo's contrast, whereas CG might render a realistic scene from geometric primitives.
- **Image processing vs Machine vision:** Machine vision (or computer vision) uses image processing techniques for automated interpretation. IP might simply filter or compress an image, while machine vision applies higher-level algorithms to understand the image (e.g. object recognition, 3D reconstruction). IP can be a step within a vision system. In machine vision, tasks include feature detection and pattern recognition, often in real-time for robotics.
- **Image processing vs Video processing:** Image processing deals with single frames; video processing deals with sequences of images over time. Video processing adds temporal dimension: one must handle motion, temporal filtering, compression of time-series (MPEG). Techniques like motion estimation, object tracking, frame interpolation are unique to video. Essentially, video processing extends IP to handle the extra correlation and data in consecutive frames.

Question 3 (a)

Question: Explain histogram matching. (10 marks)

Answer: Histogram matching (also called histogram specification) adjusts the pixel intensities of one image so that its histogram matches a specified histogram (often that of another image). The steps are:

1. Compute the cumulative distribution function (CDF) of the source image and of the target histogram.
2. For each pixel intensity r in source, find its CDF value $C_s(r)$.
3. Find the intensity z in the target such that $C_t(z) \approx C_s(r)$.
4. Map $r \rightarrow z$.

This remapping forces the output image to have the target histogram shape. It's used to standardize images or to apply a known look (like making photos have the same contrast distribution). Unlike equalization (which maps to uniform histogram), histogram matching can produce any desired histogram. It equalizes the image relative to the target.

Question 3 (b)

Question: Explain image degradation model and explain any noise model in detail. (10 marks)

Answer: The **image degradation model** expresses how an original image f is transformed into a degraded observed image g . Typically $g = f * h + \eta$, where h is a blurring kernel (point spread function) and η is additive noise. For example, motion blur can be modeled by a linear kernel h , and sensor noise by white noise η . This model assumes linear, shift-invariant blur plus noise. In general, $g(x, y) = \int f(u, v)h(x - u, y - v) du dv + \eta(x, y)$.

One **noise model: Additive Gaussian noise**. Here $\eta(x, y)$ is a sample from a Gaussian distribution $N(0, \sigma^2)$ at each pixel, independent of others. This models many natural noise processes (thermal noise, electronic noise). The probability $p(\eta) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\eta^2/(2\sigma^2)}$. Gaussian noise is easy to handle mathematically (e.g. Wiener filtering). Another is **salt-and-pepper noise** (impulse noise), where pixels randomly become black or white. Or **speckle noise** (multiplicative gamma distribution), common in radar. Each model has appropriate filtering strategies (median filter for salt-pepper, Wiener for Gaussian).

Question 4 (a)

Question: Draw block diagram of supervised learning phases. (7 marks)

Answer: See the block diagram provided in 2023 Q4(a). (Supervised learning phases: Data → Feature extraction → Train model → Prediction.)

Question 4 (b)

Question: Write short notes on: (i) Feature vector, (ii) Feature space, (iii) Decision Region, (iv) Hyperplanes and (v) Hypersurfaces. (10 marks)

Answer:

- **Feature Vector:** A numerical vector that represents an object or image, comprised of measured attributes (features). E.g. for a pixel, a feature vector could be $[R, G, B]$ or $[x, y, \text{intensity}]$. In vision, a feature vector for a region might be $[\text{avg intensity, texture, shape descriptors, ...}]$. It is the input to classifiers.

- **Feature Space:** The multidimensional space whose axes correspond to features. Each data point is a point in this space. For example, 2 features define a 2D plane, 3 features define 3D space. Patterns like clusters or class boundaries are defined in feature space.
- **Decision Region:** For a classifier, a decision region is the subset of feature space assigned to a particular class. E.g. a linear classifier has a decision region for class 1 being one side of the hyperplane.
- **Hyperplanes:** In d -dimensional space, a hyperplane is a flat affine subspace of dimension $d - 1$. It can be defined by $w^T x + b = 0$. In classification, hyperplanes arise from linear discriminant functions and separate decision regions linearly.
- **Hypersurfaces:** More general boundaries in feature space (can be curved). The decision boundary can be a hypersurface of any shape (quadratic surfaces, etc.). In nonlinear classifiers, hypersurfaces separate classes.

Each term is about representing data and decisions in the feature-attribute space.

1 9 digitizn.dvi

https://www.princeton.edu/~cuff/ele201/kulkarni_text/digitizn.pdf

2 6 Analog Image Processing Vs Digital Image Processing - GeeksforGeeks

<https://www.geeksforgeeks.org/electronics-engineering/analog-image-processing-vs-digital-image-processing/>

3 Image file format - Wikipedia

https://en.wikipedia.org/wiki/Image_file_format

4 5 Discrete Fourier transform - Wikipedia

https://en.wikipedia.org/wiki/Discrete_Fourier_transform

7 Salt-and-pepper noise - Wikipedia

https://en.wikipedia.org/wiki/Salt-and-pepper_noise

8 Minimum Euclidean Distance - an overview | ScienceDirect Topics

<https://www.sciencedirect.com/topics/engineering/minimum-euclidean-distance>

10 Point Operations - Contrast Stretching

<https://homepages.inf.ed.ac.uk/rbf/HIPR2/stretch.htm>

11 12 RGB color model - Wikipedia

https://en.wikipedia.org/wiki/RGB_color_model

13 14 25 26 Difference between K means and Hierarchical Clustering - GeeksforGeeks

<https://www.geeksforgeeks.org/machine-learning/difference-between-k-means-and-hierarchical-clustering/>

15 Chromaticity - Wikipedia

<https://en.wikipedia.org/wiki/Chromaticity>

16 23 24 What is HSI Color Space? - GeeksforGeeks

<https://www.geeksforgeeks.org/electronics-engineering/what-is-hsi-color-space/>

17 18 uomus.edu.iq

https://uomus.edu.iq/img/lectures21/MUCLecture_2024_10642263.pdf

19 20 Bayes Classification - GeeksforGeeks

<https://www.geeksforgeeks.org/machine-learning/bayes-classification/>

21 22 CMY color model - Simple English Wikipedia, the free encyclopedia

https://simple.wikipedia.org/wiki/CMY_color_model